

FINAL REPORT

Undergraduate Research Opportunities Program 1 (UROP1010)

Student name:	Nguyen Quy Tu	Date:	19/6/2026
Student ID:	V202502163	Year:	1
College:	CECS	Program:	Data Science
Faculty supervisor name:	ZengChang Qin		

Research Project Title:

Evaluating LLM-based Trading Agents in a Simulated Continuous Double Auction Market

I. Project Overview (20 points)

Field: Computational Economics & AI Agent-Based Market Simulation

Goal: Develop a reproducible framework using TBSE (Threaded Bristol Stock Exchange) to design, deploy, and evaluate AI-driven trading agents in microeconomic market environments.

Key Objectives:

- Build an automated 8-step pipeline (config → run → analyze → visualize) for systematic experimentation.
- Integrate AI-based trading agents (e.g., learning or adaptive agents) alongside traditional strategies (ZIP, ZIC, GVWY).
- Measure market efficiency using allocative efficiency and related performance metrics.
- Compare AI agents with classical rule-based traders under controlled conditions.
- Analyze price dynamics (exploration → equilibrium → stability) in the presence of intelligent agents.
- Examine the trade-off between profit, trading volume, and adaptability of AI strategies.

Significance:

- Provides a reproducible workflow for studying AI applications in microeconomic markets.
- Explores how intelligent agents interact with and potentially outperform traditional trading strategies.
- Bridges microeconomic theory with modern AI-driven market behavior (learning, adaptation, price discovery).

Summary: A structured and repeatable experimental framework for investigating how AI agents influence market dynamics, efficiency, and strategic interactions in simulated microeconomic environments.

II. Progress Update (50 points)

Task 1: Understanding the market environment and selecting the simulation framework

- Studied continuous double auction (CDA) market structure and how it supports price discovery in modern trading systems.
- Compared TBSE/BSE-style simulation with ABIDES and selected ABIDES for the final study because it supports event-driven execution, latency modeling, private messaging, and a built-in adaptive benchmark agent (HBL).
- Defined the simulation setup around a small, controlled market so that changes in agent behavior could be observed clearly and repeated consistently.

Task 2: Designing the trading agents and experiment pipeline

- Developed the agent setup for the project, including a locally hosted open-source LLM agent (Gemma 3 4B Instruct), benchmark HBL agents, and zero-intelligence liquidity agents.
- Designed structured LLM outputs so each agent can produce both reasoning text and machine-readable JSON orders.
- Added a memory module so the LLM can retain useful past trading information such as previous fills and profit-and-loss outcomes.
- Planned an experimental pipeline covering prompt structure, memory length, information access, latency, market shocks, and communication effects.

Task 3: Running pilot experiments and analyzing early results

- Completed two pilot runs: a baseline HBL/ZI market and an LLM-inclusive market.
- In the baseline run, spreads were generally tight, trading volume was steady, and prices stayed relatively calm, although the mid-price followed the fundamental value only loosely.
- In the LLM-inclusive run, trading activity increased substantially and the market became more active, but spreads were wider and price tracking did not improve clearly.
- These pilot results suggest a trade-off: the LLM increases market activity and volume, but it does not yet show stronger price efficiency than the benchmark.
- The main next steps are to run more seeds, add PnL measurement, and quantify price tracking with a numerical error metric rather than only qualitative observation.

III. Self-Evaluation (30 points)

Please assess your performance during the project based on the following evaluation criteria on a scale of 1 to 5 and make specific comments in each category.

(1-Unsatisfactory performance; 2-Below expectation; 3-Average performance; 4-Good performance; 5-Excellent performance)

Criteria	1	2	3	4	5	Comment
1 - I have maintained professionalism during the program (professional presence, punctuality, etc.)				X		
2 - I have been able to complete given tasks with timeliness, accuracy, quality, and efficiency.					X	
3 - I have demonstrated analytical thinking, critical thinking, and problem-solving skills while completing given tasks.				X		
4 - I have proposed new ideas, creative initiatives and/or applied innovative methods/ways to complete given tasks.				X		
5 - have demonstrated effective communication and collaboration with peers and supervisors (listen sincerely, be willing to help, be trusted, etc.)				X		

What are your main strengths?

- Decent logical abilities and interpretation skills.
- Dedicated and strong attitude when pursuing a goals.
- Being a reflective person, continuously looking for improvements

What are your main areas of improvement?

- Time Management Skills
- Communication and Socializing skills